Etienne Mueller

(address hidden in web version)
etiennemueller.com | contact@etiennemueller.com | github.com/EtienneMueller | (hidden)

Education

Ph.D. in Computer Science (Artificial Intelligence) 2023

Technical University Munich, Germany

- Thesis on the conversion of conventional to spiking neural networks for energy-efficient neuromorphic computation
- Research of novel biologically-inspired approaches for convolutional, recurrent and transformer architectures for natural language processing and pattern recognition with varying large datasets

M.Sc. in Product Development (Autonomous Driving) 2017

Technical University Hamburg, Germany

- Thesis on the development of autonomous racing vehicles to sense and act under time constraints
- Awarded Best Autonomous Design at Formula Student Germany
- Awarded the Incentive Prize of the Technical University of Hamburg endowed with 1,500€

Semester Abroad 2016

Institut Catholique d'Arts et Métiers Nantes, France

- Thesis on designing tools for workflow optimization of a waste utilization plant using CAD

B.Sc. in Mechanical Engineering (Robotics) 2014

Technical University Hamburg, Germany

- Thesis on measuring and increasing the accuracy and repeatability of industrial robots using MATLAB

Professional Experience

Assistant Professor AI & Data Science 2025 – Present

Carnegie Mellon | KMITL (CMKL), Thailand

- Investigating hierarchical spiking neural networks for cognitive neuromorphic computing, focusing on bio-inspired multi-timescale architectures for compositional reasoning with planned validation on various neuromorphic hardware
- Developing the curriculum for *Neural Networks and Deep Learning*, *LLMs and Agentic AI*, and *Big Data Computing*

Postdoctoral Researcher AI 2023 – 2025

University of Melbourne, Australia

- Developed neural network growth algorithms to create biologically-inspired memory cells for more efficient recurrent neural networks, implementing ML simulations of brain imaging data across different developmental stages using JAX
- Ran deep convolutional neural networks on a Slurm-based HPC with up to 4xH100 GPUs per node for automated segmentation of synchrotron brain imaging data, reducing the need for manual annotation by a factor of 10
- Attended *Nengo Summer School 2023* at University of Waterloo, Canada, on large-scale brain modelling using Nengo to program neuromorphic hardware systems and modelling cognitive and behavioural data

AI Engineer 2022 – 2023

Flowers Software, Germany

- Established the AI research department at a seed-financed startup, deploying deep learning infrastructure from scratch
- Developed a TensorFlow-based information extraction workflow on AWS to identify recurring positions on invoices that complies with EU data privacy law, saving the company over €5,000/month by eliminating third-party API costs

Technical Advisor 2021 – 2022

Technical University of Munich

- Led the technical coordination of a pilot case for an EU-funded (Horizon 2020) project (SHOP4CF)
- Creation of modular tools for autonomous factories for Industry 4.0

AI Researcher 2018 – 2022

Infineon Technologies, Germany

- Research in neuromorphic computing and spiking neural networks, leading to 11 first- and second-author publications
- Developed a TensorFlow-based toolbox for converting conventional to spiking neural networks, which was subsequently used in a research project to reduce the simulation time of hardware components in neuromorphic systems by half

Co-founder & CEO 2015 – 2016

Slive Technologies, Germany

– Developed smart wearable devices and location-based algorithms for hands-free data use in industrial environment
– Secured the Nissen Foundation Start-Up Grant (€3,000) to support early-stage product development and business growth

Formula Student Member @ e-gnition Hamburg e.V. 2012 – 2017

Developer Driverless Actuator Technology (2016 – 2017)
– 1st Place Formula Student Driverless: Autonomous Design
– 3rd Place Formula Student Driverless: Overall

Division Manager Business Plan (2014 – 2016)
– Special Award for educational video "How to Business Plan" at Formula Student Hungary

President & Team Captain (2013 – 2014)
– Special Award for Ecological Design by Magna Steyr

Division Manager Aerodynamics (2012 – 2013)

Teaching Experience

Lecturer in Cognitive Systems 2019 – 2021

– Covering topics such as cognition, biological inspired computational models and neurorobotics
– Creating and grading of exams for over 400 students

Thesis Supervision in Deep Learning and Spiking Neural Networks 2018 – 2021

– Master Thesis (2021): Performance of Time to First Spike Encoded Spiking Neural Networks
– Research Internship (2021): Conversion of Analog to Spiking Transformer Networks
– Master Thesis (2021): Conversion of Analog LSTM-based Recurrent Neural Networks
– Master Thesis (2021): Conversion of Analog GRU-based Recurrent Neural Networks
– Research Internship (2020): Carla as Open Source Platform for Analyzing and Evaluating Autonomous Driving
– Master Thesis (2020): Converting Analog to Spiking Convolutional Neural Networks for Object Detection
– Master Thesis (2019): Semantic Segmentation of Integrated Circuit Layout Images

Publications

E. Mueller, W. Qin, "Reverse Engineering Neural Connectivity: Mapping Neural Activity Data to Artificial Neural Networks for Synaptic Strength Analysis," in *8th International Conference on Information Technology (InCIT),* Chonburi, Thailand and Kanazawa, Japan, 2024, pp. 451-454.

E. Mueller, "Optimizing the Conversion of Continuous-Valued Networks to Spiking Neural Networks," *Doctoral Thesis,* Technical University of Munich, Germany, 2024, .

E. Mueller, S. Klimaschka, D. Auge, A. Knoll, "Neural Oscillations for Energy-Efficient Hardware Implementation of Sparsely Activated Deep Spiking Neural Networks," in *Association for the Advancement of Artificial Intelligence (AAAI) Practical DL,* Online (Vancouver, Canada), 2022, pp. 1-7.

E. Mueller, D. Auge, A. Knoll, "Exploiting Inhomogeneities of Subthreshold Transistors as Populations of Spiking Neurons," in *International Conference on Natural Computation, Fuzzy Systems and Knowledge Discovery (ICNC-FSKD),* Online (Fuzhou, China), 2022, pp. 1-8.

E. Mueller, V. Studenyak, D. Auge, A. Knoll, "Spiking Transformer Networks: A Rate Coded Approach for Processing Sequential Data," in *7th Int. Conference on Systems and Informatics (ICSAI),* Online (Jiaxing, China), 2021, pp. 1-5.

E. Mueller, J. Hansjakob, D. Auge, A. Knoll, "Minimizing Inference Time: Optimization Methods for Converted Deep Spiking Neural Networks," in *International Joint Conference on Neural Networks (IJCNN),* Online (Shenzhen, China), 2021, pp. 1-8.

D. Auge, J. Hille, **E. Mueller**, A. Knoll, "A Survey of Encoding Techniques for Signal Processing in Spiking Neural Networks," *Neural Processing Letters,* vol. 53, issue 6, pp. 4693-4710, Dec 2021.

E. Mueller, D. Auge, A. Knoll, "Normalization Hyperparameter Search for Converted Spiking Neural Networks," in *Bernstein Computational Neuroscience Conference,* Online (Berlin, Germany), 2021, P 8.

D. Auge, J. Hille, **E. Mueller**, A. Knoll, "Hand Gesture Recognition in Range-Doppler Images Using Binary Activated Spiking Neural Networks," in *IEEE International Conference on Automatic Face and Gesture Recognition,* Online (Jodhpur, India), 2021, pp. 1-7.

D. Auge, J. Hille, F. Kreutz, **E. Mueller**, A. Knoll, "End-to-end Spiking Neural Network for Speech Recognition Using Resonating Input Neurons," in *30th International Conference on Artificial Neural Networks (ICANN),* Online (Bratislava, Slovakia), 2021, pp. 245-256.

E. Mueller, J. Hansjakob, D. Auge, "Faster Conversion of Analog to Spiking Neural Networks by Error Centering," in *Bernstein Computational Neuroscience Conference,* Online (Berlin, Germany), 2020, P 146.

D. Auge, P. Wenner, **E. Mueller**, "Hand Gesture Recognition using Hierarchical Temporal Memory on Radar Sequence Data," in *Bernstein Computational Neuroscience Conference,* Online (Berlin, Germany), 2020, P 3.

Daniel Auge, **E. Mueller**, "Resonate-and-Fire Neurons as Frequency Selective Input Encoders for Spiking Neural Networks," Chair of Informatics, TUM, Munich, Technical Report TUM-I2083. 2020

Presentations

Sustainability Expo 2025 Oct. 2025
Conference Talk

International Conference on Information Technology (InCIT) Nov. 2024
Conference Talk

Bioinformatics Meetup Aug. 2024
Talk at Melbourne Bioinformatics

International Conference on Neuromorphic Computing & Engineering (ICNCE) Jun. 2024
Conference Poster Presentation

Doktorandenhuette Mar. 2022
Talk at the Department of Robotics, Artificial Intelligence and Real-time Systems, TUM

Association for the Advancement of Artificial Intelligence (AAAI) Feb. 2022
Conference Talk

International Conference on Systems and Informatics (ICSAI) Nov. 2021
Conference Talk

Bernstein Computational Neuroscience Conference Sep. 2021
Conference Poster Presentation

International Joint Conference on Neural Networks (IJCNN) Jul. 2021
Conference Talk

Doktorandenhuette Nov. 2020
Talk at the Department of Robotics, Artificial Intelligence and Real-time Systems, TUM

Infineon InnoWeek Oct. 2020
Internal Research Conference Presentation

Bernstein Computational Neuroscience Conference Poster Oct. 2020
Conference Poster Presentation

Ph.D. Second Year Presentation Jul. 2020
Public Presentation

Infineon Deep Learning Symposium Apr. 2019
Conference Poster Presentation

Doktorandenhuette Dec. 2018
Talk at the Department of Robotics, Artificial Intelligence and Real-time Systems, TUM

Open Source Projects

Connectogenesis

- A python library to analyse growth of zebrafish larvae using selective plane illumination microscopy brain imaging data
- Tools and scripts for mapping neural images from different days to each other, converting neural activity to artificial neural networks, and training for synaptic strength analysis

Convert2SNN

- A TensorFlow-based library that converts conventionally trained neural networks with continuous activation functions to spiking neural networks, with minimal to no performance loss, to estimate energy consumption in neuromorphic systems
- Supports key spike encoding techniques, including rate, population, and temporal coding, with the ability to estimate spike counts for efficiency evaluation, reducing the need for extensive hardware simulations during development

Skills

Programming Python, TensorFlow, JAX, Bash, Swift, Xcode, C++, SQL
Languages German (native), French (native), English (fluent), Spanish (basic), Chinese (basic)
Interests Music (Piano, Ukulele), Fitness, Health